United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Mid Exam Spring 2026

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: **30**   Duration: 1 hour 30 Minutes

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

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1. (a) Show the corresponding inputs and outputs from **Lexical analyzer** to **Code Generator** phase of compiler based on the following expression where a and b are **int** types and c is **float** type. 4

$$a = a/2 + (b * c)^\wedge 5$$

(b) A large **study table** cannot fit into a small box in its assembled form. To solve this, Lisa decides to disassemble the table into smaller parts so that they can be packed easily. However, she becomes concerned about how to remember the correct order in which the parts should be put back together later. Observing Lisa's approach, Sam is reminded of a particular phase (denoted as XYZ) in the language compilation process that he studied in his compiler course. 3

Identify the role of **XYZ** in the compilation process. In your answer, describe how this phase works and explain how Lisa's situation is similar to this phase.

c) A programmer writes a complex arithmetic expression in a high-level programming language. Before converting it into machine-level instructions, the compiler transforms the expression into a simpler, structured representation using some identifiers. This representation is easier to analyze and can be used across different machine architectures. 3

Identify the compiler's phase being described. Explain how this phase works and why such a representation is useful in the compilation process and also give an example.

2. Write down the three address representation of the following expression. 4

$$\mathbf{P = (a+ (b \wedge c - d / (2 * a)^\wedge 6) / 2 * 3) + (2^\wedge e \% 2/5) / 2^\wedge 3 / a^\wedge b^\wedge c}$$

3. Draw Syntax Tree for the following expression. 4

$$\mathbf{x = u + 2 * 9 / 2 / v \wedge (2 - (3^\wedge t^\wedge 2) - 2 / (5 \wedge 6)^\wedge 9 * 2) / 2 - r / p}$$

4. Prove that the following expression **“(x + 2*y) - (3*z + 1)”** can be represent by the grammar given below. 4

$S \rightarrow S + S \mid S - S \mid (S) \mid T$
$T \rightarrow X * X \mid X \% X \mid X$
$X \rightarrow x \mid y \mid z \mid Y$
$Y \rightarrow 0 \mid 1 \mid 2 \mid 3$

5. Convert the following infix expression to a prefix expression. 4

$$\mathbf{(2/7\text{^}(x \% z / y + r)\text{^}2\text{^}1) - r/p\text{^}(6 - 3/9)\text{^}p \% t + 1\text{^}3}$$

6. Evaluate the following postfix expression. Here, a = 3, b = 5 4

2 a 2 2 ^ a / + 2 (10) 2 ^ / - 3 b ^ 2 + 3 6 2 ^ + * - +